

SurfaceLayerDiffusivity: first results

First physical comparison | 21 September 2026 | preliminary, diagnostic exclusions apply.

Question: can transport confined to one or two near-wall faces improve coarse LES? Four paired GABLS1 runs use WENO9 at 12.5 m resolution (32^3 cells), a 400 m cube, the same frozen source, and 9 h of integration.

Finding: all three SurfaceLayerDiffusivity variants move the mean u , mean θ and w^2 profiles farther from the fixed 1 m LES median by RMS error over native levels below 200 m. This result applies to this grid and these settings; it is not a general verdict on the closure.

Near-wall response: at 12.5 m, final-hour w^2 falls by 85–93%. Skewness changes from +0.56 in the control to -0.21, -0.25 and -0.42. Domain-integrated resolved TKE falls by 15–20%. The first-face variance is suppressed much more strongly than the integrated energy.

Surface exchange: final-hour u^* rises from 0.279 m/s to 0.292, 0.289 and 0.301 m/s, versus a reference mean of 0.262 m/s. Surface cooling also strengthens, moving farther from the reference. One-face 100 and 300 s responses are similar; two-face support increases local suppression.

Sensitivity: the preceding hour shows the same direction of near-wall variance, skewness, surface-drag and resolved-energy changes. These are single-seed comparisons, not uncertainty estimates. The LES reference median is not observational truth; w^3 has no direct archive reference.

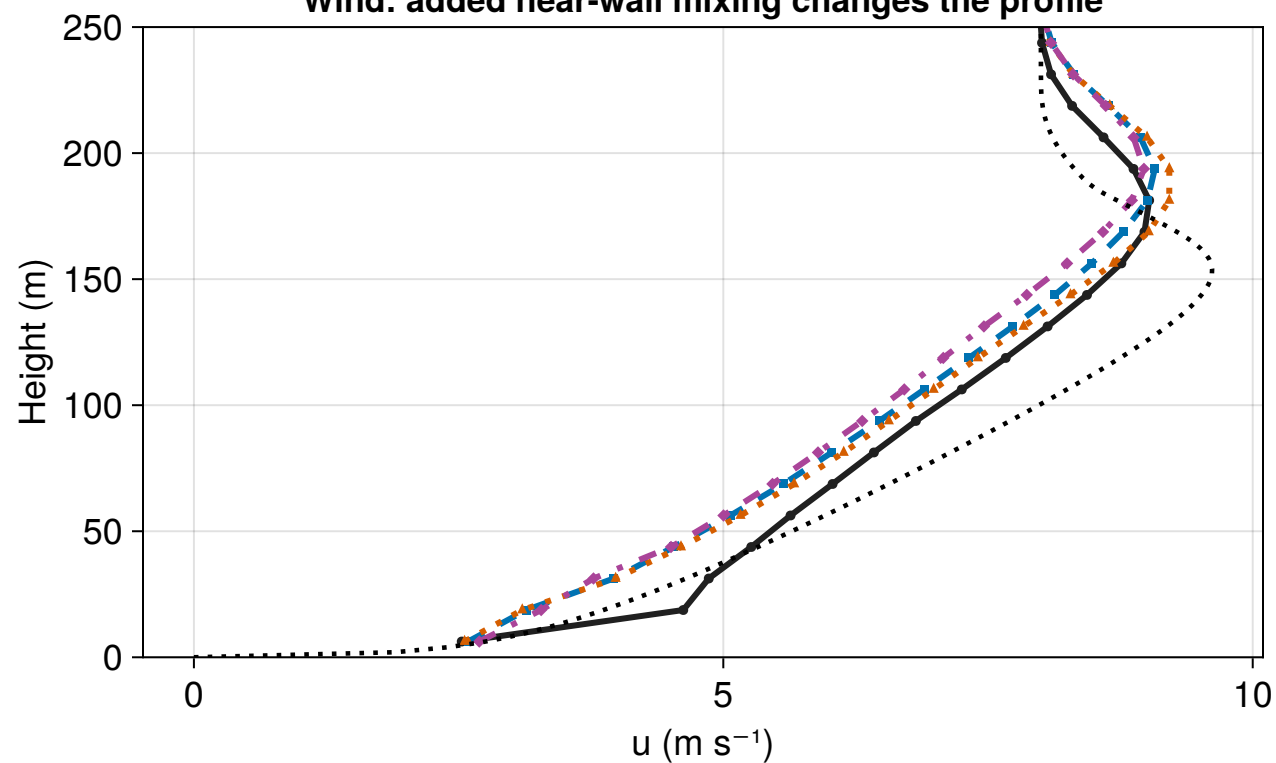
Diagnostic exclusion: SGS flux was incorrectly reported as zero because the generic diagnostic omits the implicit contribution. Combined flux, stress-based depth and dependent budgets are withheld. Mean fields, resolved moments/energy and direct wall exchange remain usable. The model does apply implicit transport. Correct transport diagnostics are the next validation step.

Reading the figures: native-height profiles are true 8–9 h means; timelines use instantaneous output. Colors, line styles and markers distinguish cases. Exact data, Julia source, reference hashes and diagnostic exclusions accompany this brief; all original DYCOMS/GABLS1 report material is retained.

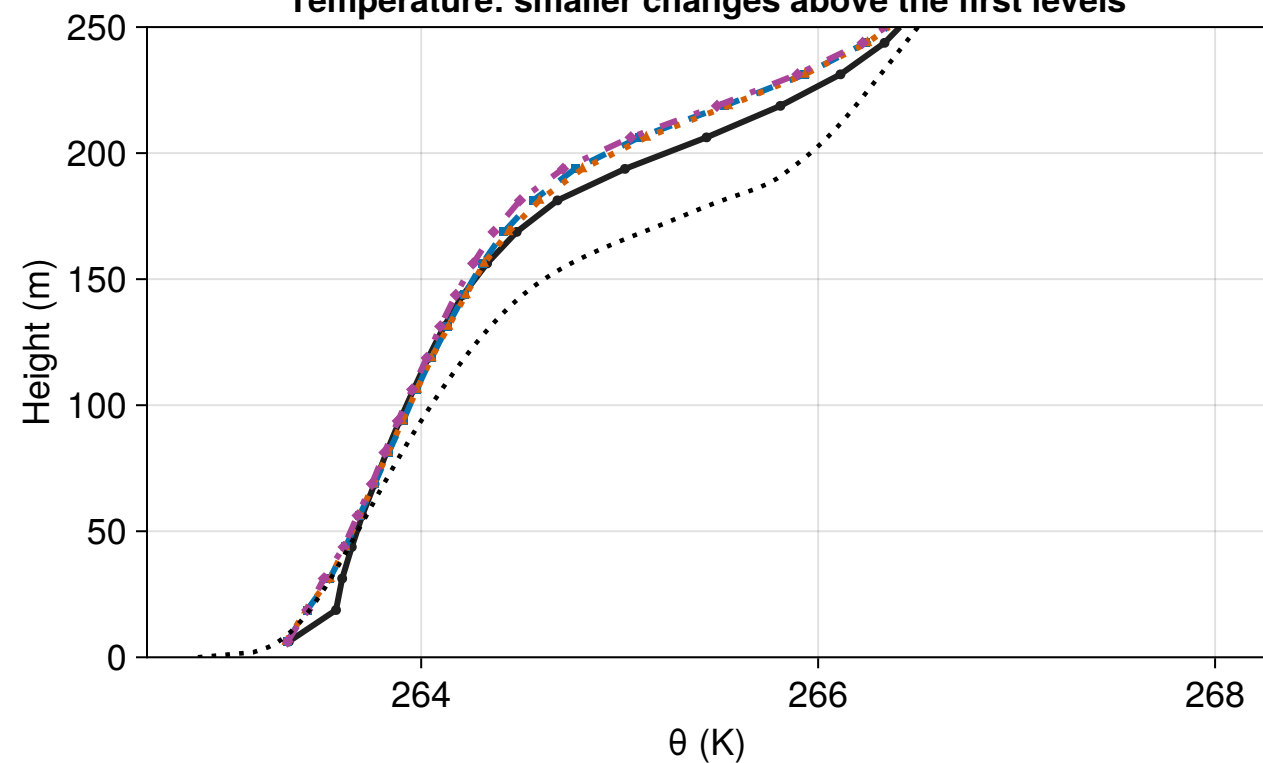
Strong response; no improvement in this test

GABLS1 · WENO9 · 12.5 m grid (32^3) · final hour, 8–9 h · same source and paired seed

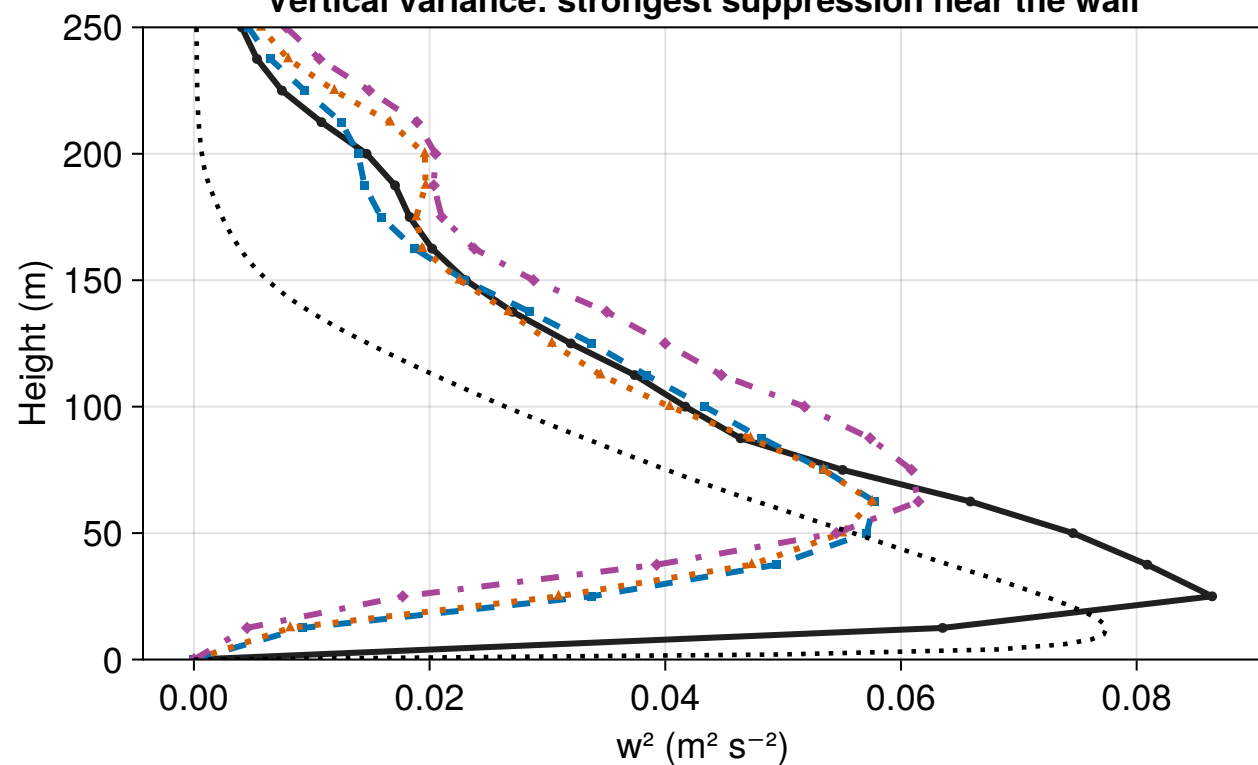
Wind: added near-wall mixing changes the profile



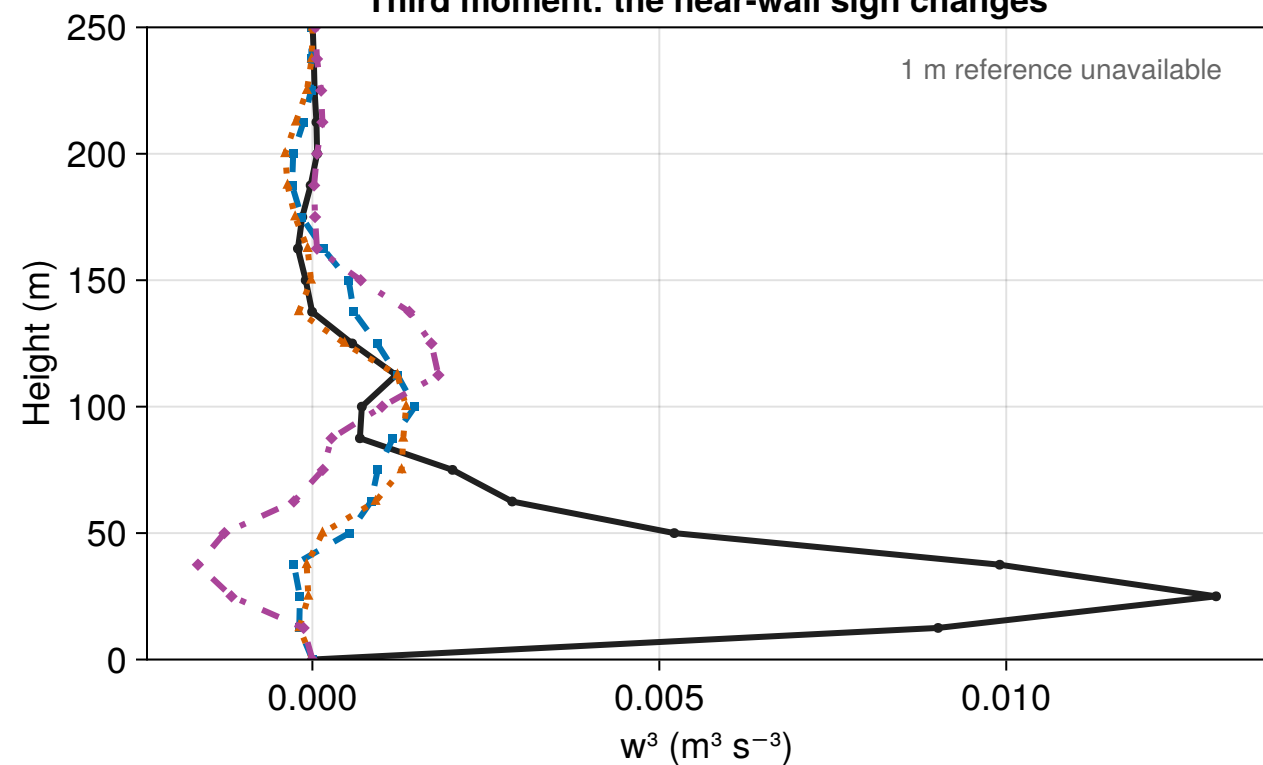
Temperature: smaller changes above the first levels



Vertical variance: strongest suppression near the wall



Third moment: the near-wall sign changes

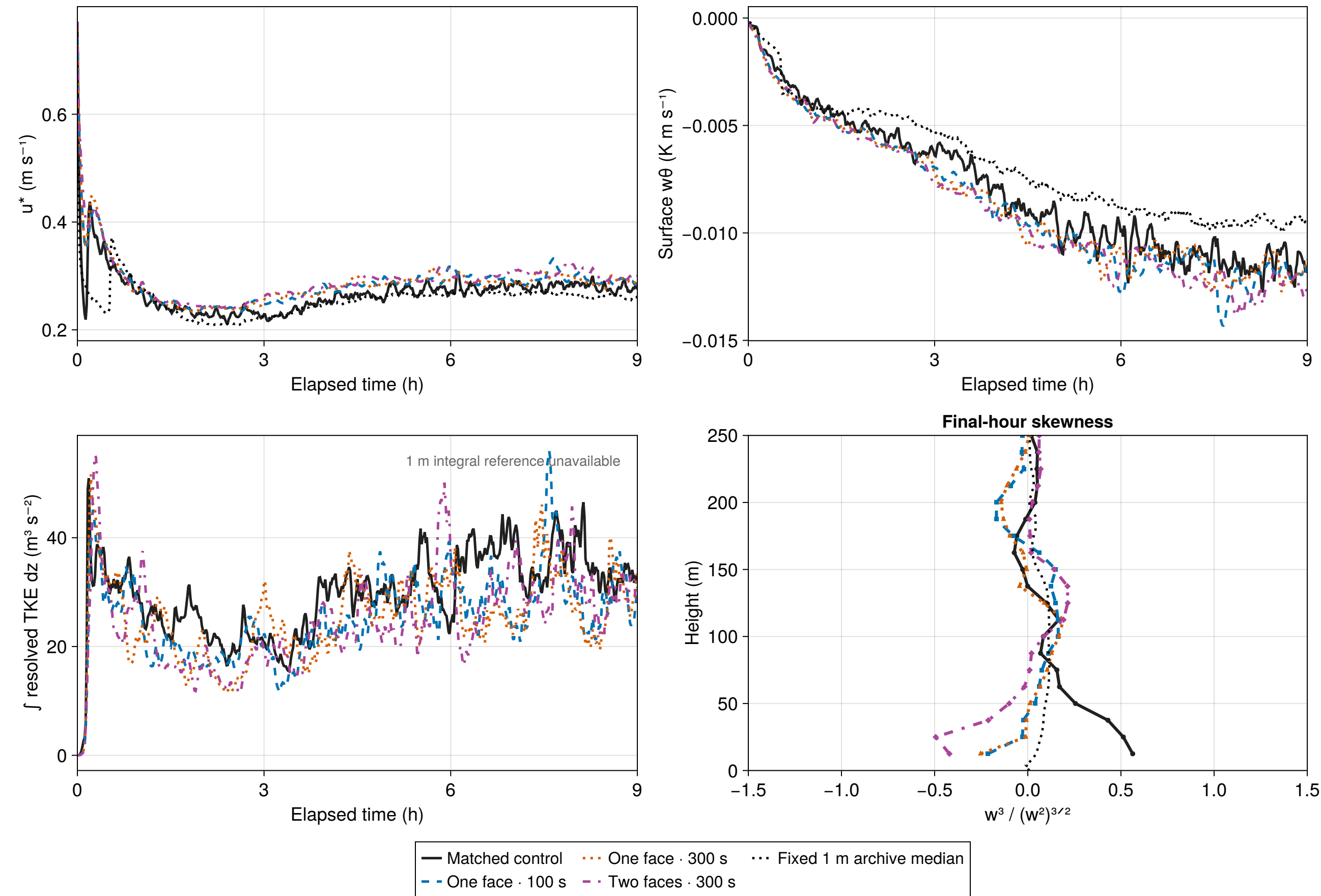


— Matched control - - - One face · 300 s ··· Fixed 1 m archive median
- · - One face · 100 s - - - Two faces · 300 s

Native simulation levels; black dotted reference is an LES ensemble median, not observational truth.
Single seed. SGS/combined flux, stress-based depth and dependent budgets are withheld pending a diagnostic correction.

Follow the response beyond the mean profile

Surface exchange through 9 h; skewness from the final-hour native-face moments



Surface flux is obtained from the wall boundary condition and remains usable. Resolved TKE excludes SGS energy. Skewness is a ratio of window-mean moments; omit model levels with $w^2 < 10^{-6} \text{ m}^2 \text{s}^{-2}$. No uncertainty band is inferred.